Chapter 2

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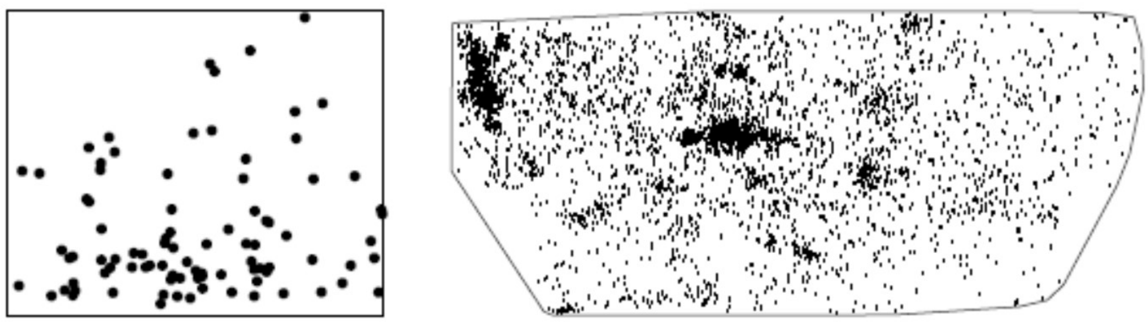
Spatial Point Pattern

Background, Definition, Application Areas and SPP Data Types

Background, Definition

- A 'spatial point process' is a stochastic process in which we observe spatial locations of some things or events of interest within a bounded region A.
- The locations of the events generated by a point process in the area of study A will be called a point pattern.
- The spatial arrangement of points is the main focus of investigation.



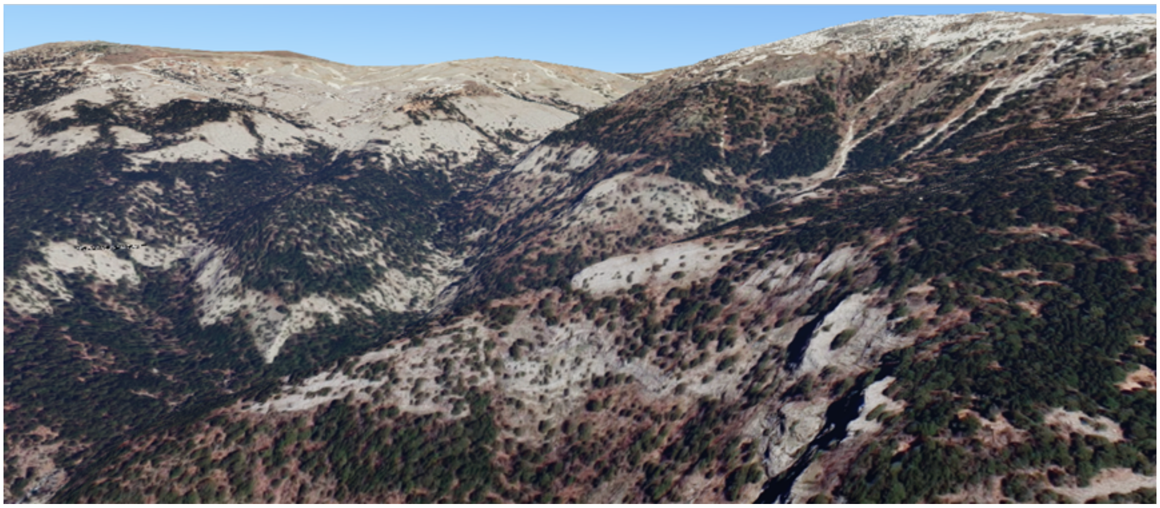
Ref: [3] page 3. Figure 1.1

Application Areas

Examples include:

- the locations of trees in a forest, the spatial distribution of the trees, competition or coordination between different species of vegetation
- gold deposits mapped in a geological survey,
- stars in a star cluster,
- road accidents,
- earthquake epicentres,
- mobile phone calls,
- facility locations,
- animal sightings,
- cases of a rare disease, are they clustered, reasons why,
- place where crime occurs – hotspots (we need good data for this: check out:

Example 1: Trees



Example 2: Cholera

Example 3: Stars – Ara Constellation

Ref: https://www.constellation-guide.com/constellation-list/ara-constellation/ Ara was one of the 48 Greek constellations listed by the astronomer Claudius Ptolemy in the 2nd century. Ara contains two stars brighter than magnitude 3.00 and three stars located within 10 parsecs (32.6 light years) of Earth.

Example 4: Clinics

Download the data from the following:

https://web1.capetown.gov.za/web1/OpenDataPortal/DatasetDetail?DatasetName=Clinics

```r
library(sf)
```

```
Warning: package 'sf' was built under R version 4.4.3
```

```
Linking to GEOS 3.13.0, GDAL 3.10.1, PROJ 9.5.1; sf_use_s2() is TRUE
```

```r
RSA_roads = "C:/Users/01438475/OneDrive - University of Cape Town/ASDA/DataSets/zafrds8ff5a/Z
  st_read() |> st_transform(4326)
```

```
Reading layer `ZAF_roads' from data source
  `C:\Users\01438475\OneDrive - University of Cape Town\ASDA\DataSets\zafrds8ff5a\ZAF_roads.s
  using driver `ESRI Shapefile'
Simple feature collection with 5559 features and 5 fields
Geometry type: MULTILINESTRING
Dimension:     XY
Bounding box:  xmin: 16.48784 ymin: -34.82747 xmax: 32.85267 ymax: -22.17693
Geodetic CRS:  WGS 84
```

```r
RSA_biome = "C:/Users/01438475/OneDrive - University of Cape Town/ASDA/DataSets/rsabiome4xkhi
  st_read() |> st_transform(4326)
```

```
Reading layer `RSA_biome' from data source
  `C:\Users\01438475\OneDrive - University of Cape Town\ASDA\DataSets\rsabiome4xkhi\RSA_biome
  using driver `ESRI Shapefile'
Simple feature collection with 3127 features and 1 field
Geometry type: POLYGON
Dimension:     XY
Bounding box:  xmin: 16.45296 ymin: -34.8336 xmax: 32.8928 ymax: -22.12583
Geodetic CRS:  Hartebeesthoek94
```

```r
clinics_sf = "C:/Users/01438475/OneDrive - University of Cape Town/ASDA/DataSets/Clinics/SL_C
  st_read() |> st_transform(4326)
```

```
Reading layer `SL_CLNC' from data source
  `C:\Users\01438475\OneDrive - University of Cape Town\ASDA\DataSets\Clinics\SL_CLNC.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 149 features and 5 fields
Geometry type: POINT
Dimension:     XY
Bounding box:  xmin: 18.34268 ymin: -34.19491 xmax: 18.90847 ymax: -33.51262
Geodetic CRS:  WGS 84
```

```r
ct.wards_sf <- "C:/Users/01438475/OneDrive - University of Cape Town/ASDA/DataSets/sa/CPT/ele
  st_read(quiet = TRUE) |>
  st_set_crs(4326)

st_crs(ct.wards_sf)
```

```
Coordinate Reference System:
  User input: EPSG:4326
  wkt:
GEOGCRS["WGS 84",
    ENSEMBLE["World Geodetic System 1984 ensemble",
        MEMBER["World Geodetic System 1984 (Transit)"],
        MEMBER["World Geodetic System 1984 (G730)"],
        MEMBER["World Geodetic System 1984 (G873)"],
        MEMBER["World Geodetic System 1984 (G1150)"],
        MEMBER["World Geodetic System 1984 (G1674)"],
        MEMBER["World Geodetic System 1984 (G1762)"],
        MEMBER["World Geodetic System 1984 (G2139)"],
        MEMBER["World Geodetic System 1984 (G2296)"],
        ELLIPSOID["WGS 84",6378137,298.257223563,
            LENGTHUNIT["metre",1]],
        ENSEMBLEACCURACY[2.0]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    CS[ellipsoidal,2],
        AXIS["geodetic latitude (Lat)",north,
            ORDER[1],
            ANGLEUNIT["degree",0.0174532925199433]],
        AXIS["geodetic longitude (Lon)",east,
            ORDER[2],
            ANGLEUNIT["degree",0.0174532925199433]],
    USAGE[
        SCOPE["Horizontal component of 3D system."],
        AREA["World."],
        BBOX[-90,-180,90,180]],
    ID["EPSG",4326]]
```

```r
st_crs(clinics_sf)
```

```
Coordinate Reference System:
  User input: EPSG:4326
  wkt:
GEOGCRS["WGS 84",
    ENSEMBLE["World Geodetic System 1984 ensemble",
        MEMBER["World Geodetic System 1984 (Transit)"],
        MEMBER["World Geodetic System 1984 (G730)"],
        MEMBER["World Geodetic System 1984 (G873)"],
        MEMBER["World Geodetic System 1984 (G1150)"],
```

```
        MEMBER["World Geodetic System 1984 (G1674)"],
        MEMBER["World Geodetic System 1984 (G1762)"],
        MEMBER["World Geodetic System 1984 (G2139)"],
        MEMBER["World Geodetic System 1984 (G2296)"],
        ELLIPSOID["WGS 84",6378137,298.257223563,
            LENGTHUNIT["metre",1]],
        ENSEMBLEACCURACY[2.0]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    CS[ellipsoidal,2],
        AXIS["geodetic latitude (Lat)",north,
            ORDER[1],
            ANGLEUNIT["degree",0.0174532925199433]],
        AXIS["geodetic longitude (Lon)",east,
            ORDER[2],
            ANGLEUNIT["degree",0.0174532925199433]],
    USAGE[
        SCOPE["Horizontal component of 3D system."],
        AREA["World."],
        BBOX[-90,-180,90,180]],
    ID["EPSG",4326]]
```

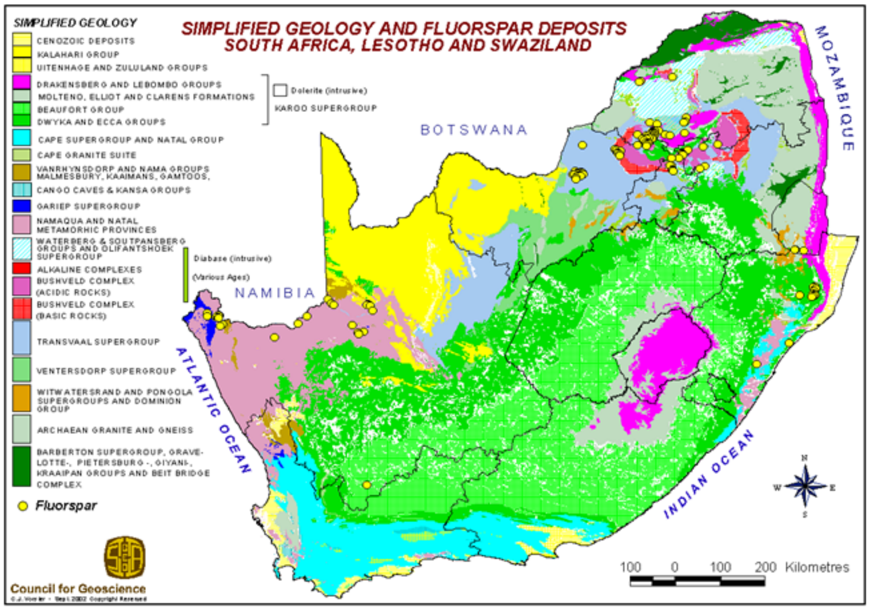
```r
class(clinics_sf)
```

```
[1] "sf"         "data.frame"
```

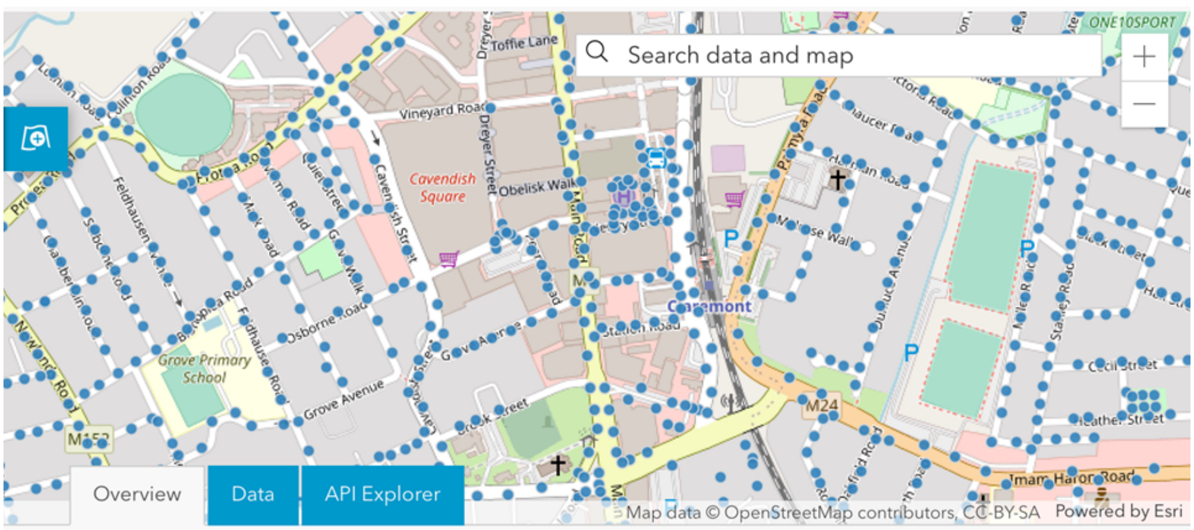
```r
summary(clinics_sf)
```

```
     LCTN                 ATHY                 NAME                CLASS          
 Length:149          Length:149          Length:149          Length:149        
 Class :character    Class :character    Class :character    Class :character  
 Mode  :character    Mode  :character    Mode  :character    Mode  :character  
     RGN                      geometry  
 Length:149          POINT        :149  
 Class :character    epsg:4326    :  0  
 Mode  :character    +proj=long...:  0  
```

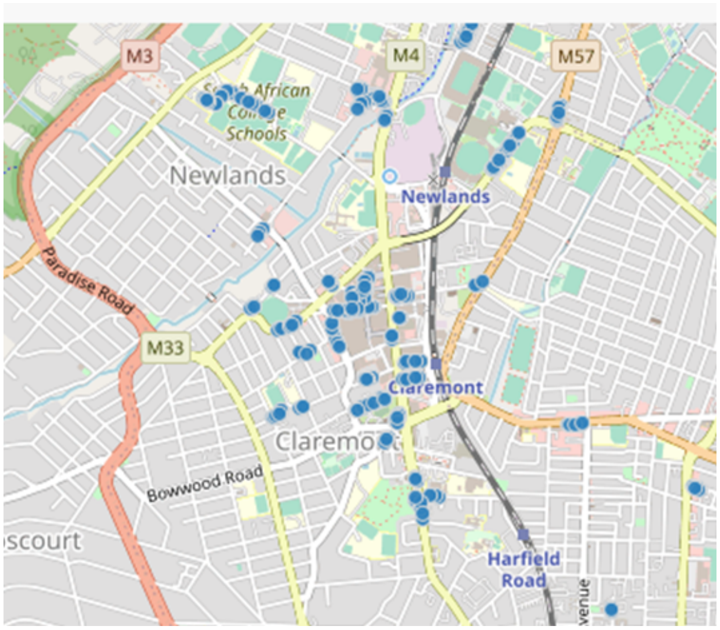
Example 5: Fluorspar



Example 6: Public Electricity Lighting



Example 7: On Street Parking



Data Resources for SPP for Cape Town Municipality

- https://odp-cctegis.opendata.arcgis.com/search?collection=Dataset
- https://odp.capetown.gov.za/datasets/wards/data
- Parks
- Wards shape file
- MyCiti Bus Stops
- Libraries

(Statistics by place)

Cape Town by documents: GeoDa Center

What is common in all these examples?

We have an event of interest (crime, trees, animals...etc.)

Point patterns are the locations of these events $\{\mathbf{s}_i\}$

Key difference from other spatial data: All points are known mapped pattern Selection bias